SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

(Deemed to be University) • Chennai

ECA1804 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO5 CASE STUDY BASED ASSIGNMENT

DWDM Metro Link Commissioning: From Field Measurements to a Standards-Based Upgrade Decision

60 km G.652.D link, 8 × 10 Gb/s to 1.6 Tb/s • (1 × 100 = 100 Marks)

Student Name	AJAY G
Register Number	192412161
Department	Electronics and Communication Engineering (ECE)
College / University	SIMATS Engineering
Academic Year	III Year (3rd Year)
Course Faculty	Princy Magdaline P
Submission Date	25 October 2026

Report workflow: REQUIREMENT → MEASUREMENT → ANALYSIS → ROOT-CAUSE IDENTIFICATION → REPAIR AND UPGRADE → VERIFICATION → DEPLOYMENT DECISION

Executive summary

The 60 km metro link was measured as-built at **22.52 dB** against a design budget of 21.25 dB, an excess of **1.27 dB**. The received power is -21.52 dBm, and after the 1.00 dB dispersion penalty and the extinction-ratio (ER) penalty the system margin is only **0.38 dB for Channel 3** and **-1.75 dB for Channel 5** against the 3 dB requirement. The link is **power- and penalty-limited, not noise-limited** (OSNR 47 dB against 20 dB required). The faults are: a poor LC/UPC patch-panel connector at 59.90 km (event 3, 1.20 dB, -22 dB reflectance, with a second connector 9 m behind it), a mechanical splice at 31.50 km (event 2, 0.62 dB) and, **specific to Channel 5 only**, a transmitter with ER 4.5 dB and DJ 0.30 UI. After the repairs and a launch-power increase to +2.5 dBm per channel, the 1.6 Tb/s upgrade uses **Option R (4 × 400G Si-photonic pluggables on 75 GHz)**.

Item	Result
As-built loss / margin	22.52 dB (design 21.25 dB); margin Ch3 0.38 dB, Ch5 -1.75 dB (needs ≥ 3 dB)
Dominant excess loss	Event 3 connector (+0.70 dB over the 0.5 dB allowance), then event 2 mechanical splice (+0.57 dB)
BER	Ch3 1.5e-14 (meets 10^{-12}); Ch5 1.3e-06 (fails by 6 decades)
Quantum limit	Direct detection 13.5 photons/bit = -47.6 dBm at 10 Gb/s, **23.6 dB** better than the -24 dBm receiver
Selected upgrade	Option R: 300 GHz, 5.33 b/s/Hz, 37.5 pJ/bit, 60 W for 1.6 Tb/s (vs 560 W with 10G SFP+)
Decision	**YES - deploy after repair, Channel 5 transmitter replacement and re-verification**

Assumptions and method

Everything is calculated from the data in the case study. Where it was not stated, the following interpretation was used and is flagged in the text.

Point	Interpretation used
OTDR one-way trace loss	Fibre + connectors + splices = 12.6 + 2.7 + 1.22 = 16.52 dB (the 6.0 dB DEMUX lies after the measured plant and is not on the trace)
ER penalty	Computed for both channels as the question asks, \u03b4$_{ER}$ = 10 log$_{10}$[(r+1)/(r-1)], r = $10^{ER/10}$. The -24 dBm sensitivity is valid only for ER ≥ 8.2 dB; Channel 3 (9.0 dB) satisfies this, so its margin without the ER penalty is also given.
OSNR	OSNR = 58 + P_{in} - NF with P_{in} = -6 dBm per channel at the booster (8 channels); a booster-only link has no in-line amplifier, so OSNR does not degrade with distance
Wavelength	1550 nm, hν = 1.28×10^{-19} J (for the quantum-limit calculation); c = 2.998×10^{8} m/s as given
Tools	The measured traces were not supplied; Figures 3.1 and 4.1 are schematic reconstructions drawn from the stated event data and spatial/dead-zone values, not field captures.

STAGE 1 - INTERPRET THE MEASUREMENT / PERFORMANCE REQUIREMENT

10 Marks | K4 - Analyze

1.1 Instruments and the quantity each one validates

Instrument	Quantity validated on this link	Limit / caution
Optical power meter + stabilised source	End-to-end insertion loss and received power per channel; per-connector loss (0.5 dB values); the final pass/fail of the 3 dB margin	Gives total loss only, not the location; needs reference cords and clean end-faces
OTDR (1550 nm)	Fibre attenuation (0.21 dB/km), splice and connector loss, reflectance, event distance, fibre length	Dynamic range, event dead zone (EDZ) and attenuation dead zone (ADZ); loss of a splice is best taken bidirectionally
Optical spectrum analyser (OSA)	Channel wavelengths on the G.694.1 grid, per-channel power, OSNR (0.1 nm), EDFA gain tilt	In-band ASE for OSNR; cannot see eye or BER
Eye-diagram / BER tester	Eye opening, ER, jitter (DJ, RJ), Q-factor and BER against 10^{-12}	Channel-by-channel; shows the symptom, not the cause
Dispersion / PMD analyser	Accumulated CD (ps/nm) and DGD (ps) of the 60 km span	Needed to confirm the 17 ps/(nm km) and 0.1 ps/√km assumptions
Fibre end-face inspection scope	Cleanliness and damage of connector end-faces (IEC 61300-3-35)	First action for any connector event

1.2 Components of a practical budget

Budget	Components that must be included
Power / loss	Launch power, fibre attenuation (αL), connectors, splices (fusion vs mechanical), DEMUX/MUX loss, repair margin (future splices), aging and temperature, system margin ≥ 3 dB
Penalties	Chromatic-dispersion penalty, extinction-ratio penalty, PMD, jitter (timing) penalty, reflection / crosstalk (OTDR reflectance values), polarisation-dependent loss
Noise / OSNR	EDFA ASE (NF, input power per channel), OSNR = 58 + P_{in} - NF in 0.1 nm, shot and thermal noise at the receiver

1.3 Eye, jitter, ER, Q and BER

The eye height is set by the ER and the receiver noise: a low ER (small gap between the 1 and 0 power levels) shrinks the vertical opening, so $Q = (I_1 - I_0)/(\sigma_1 + \sigma_0)$ falls and the BER = ½ erfc(Q/√2) rises. The eye width is set by jitter: total jitter at a given BER is TJ = DJ + n·RJ (n = 14.07 for 10^{-12}). $Q \geq 7.03$ is needed for BER ≤ 10^{-12}. A partially closed eye with high DJ and low ER (Channel 5) points to the transmitter, while a clean eye whose margin is too low (Channel 3) points to a power-budget problem.

1.4 OTDR limits

The **dynamic range** (DR) is the difference between the initial backscatter level and the noise floor and decides how far (how much loss) can be seen; the rule used here is DR ≥ one-way trace loss + 6 dB. The **event dead zone** (EDZ) is the minimum distance after a reflective event at which a second event can be seen as a separate one, and the **attenuation dead zone** (ADZ) is the minimum distance after which the loss of the second event can be measured. Wider pulses give more DR but longer dead zones (10 ns: DR 20 dB, EDZ 1 m, ADZ 4 m; 100 ns: 27 dB, 5 m, 15 m; 1 µs: 33 dB, 40 m, 100 m), so one pulse width cannot satisfy both the length of the 60 km link and the 9 m patch cord.

1.5 Standards, splices and connectors

Item	Relevance
ITU-T G.652.D	Low-water-peak single-mode fibre: attenuation ≤ 0.4 dB/km at 1550 nm (0.21 measured), D ≈ 17 ps/(nm km), PMD link design value ≤ 0.2 ps/√km; usable on the whole C-band grid
ITU-T G.694.1	DWDM frequency grid anchored at 193.1 THz with 12.5, 25, 50, 100 GHz spacing and flexible (flex-grid) slots of 12.5 GHz granularity; used in Stage 5
Fusion vs mechanical splice	Fusion: 0.02 - 0.1 dB, reflectance below -60 dB, permanent. Mechanical: 0.1 - 0.5 dB and index-gel reflection; temperature and aging sensitive. Event 2 is the 13th splice and is mechanical.
SC / LC / MPO / CS connectors	SC: 2.5 mm ferrule, push-pull, larger; LC: 1.25 mm, high density (duplex SFP+/CFP modules); MPO: multi-fibre trunk (12-24 fibres); CS: very-small-form-factor duplex for newer high-density 400G modules. UPC polish ≈ -50 dB; APC (8°) ≈ -60 dB.

1.6 Constraints

- No re-laying of fibre: the same 60 km G.652.D pair must carry 1.6 Tb/s (160 times the capacity of one 10G channel).
- System margin ≥ 3 dB at BER 10^{-12} (Q ≥ 7.03), ER ≥ 8.2 dB for the receiver sensitivity to be valid.
- Spectrum: the fibre must stay within the C-band G.694.1 grid; energy: the upgrade should minimise pJ/bit (legacy 10G SFP+ = 350 pJ/bit).

Performance Requirement Statement

The repaired and upgraded 60 km G.652.D link between the two data centres shall carry 1.6 Tb/s on the existing fibre pair, with a total optical loss no greater than 21.25 dB (every OTDR event within its allowance: splice ≤ 0.1 dB, connector ≤ 0.5 dB, no event with reflectance worse than -35 dB), a received-power margin of at least 3 dB after the dispersion and extinction-ratio penalties, a Q-factor ≥ 7.03 (BER ≤ 10^{-12}) with ER ≥ 8.2 dB and an eye width of at least 0.6 UI for every legacy channel, an OSNR margin of at least 20 dB (0.1 nm) over the upgrade option's requirement, channels on the ITU-T G.694.1 grid, and the lowest energy per bit among the standards-based options.

STAGE 2 - LINK-BUDGET, DISPERSION AND OSNR CONFIGURATION

20 Marks | K4/K5 - Analyze and Evaluate

2.1 Fibre attenuation and as-built link loss

Fibre attenuation = αL = 0.21 dB/km × 60 km = **12.6 dB**. The design loss uses the allowances; the as-built loss replaces them by the measured values. Four connectors: event 3 (1.20 dB) plus three measured at 0.5 dB (event 4 included) = 1.20 + 1.50 = 2.70 dB. Thirteen splices: twelve fusion at 0.05 dB (0.60 dB, event 1 included) plus event 2 (0.62 dB) = 1.22 dB.

Component	Design allowance	As-built (measured)	Excess
Fibre, 60 km × 0.21 dB/km	12.60 dB	12.60 dB	0.00 dB
Connectors (4)	4 × 0.5 = 2.00 dB	1.20 + 3 × 0.5 = 2.70 dB	+0.70 dB
Splices (13)	13 × 0.05 = 0.65 dB	12 × 0.05 + 0.62 = 1.22 dB	+0.57 dB
DEMUX	6.00 dB	6.00 dB	0.00 dB
Total	**21.25 dB**	**22.52 dB**	**+1.27 dB**

The as-built link is 1.27 dB worse than the design. The excess is concentrated in two events: the poor connector (event 3) contributes 1.20 - 0.50 = **0.70 dB** and the mechanical splice (event 2) contributes 0.62 - 0.05 = **0.57 dB**, together 1.27 dB, which equals the total excess. The other three connectors (0.5 dB each) and the other twelve splices (0.05 dB each) are exactly on their allowances.

2.2 Dispersion and extinction-ratio penalties

$$x = B \cdot D \cdot L \cdot \sigma_\lambda = 10\times10^{9} \times 17\times10^{-12} \times 60 \times 0.03 = 0.306$$

$$\delta_D = 5 \log_{10}[1 + 2\pi x^2] = 5 \log_{10}[1 + 2\pi(0.306)^2] = \mathbf{1.00\ dB}$$

The accumulated dispersion is 17 × 60 = 1020 ps/nm; with a 0.03 nm source this spreads the pulse by 30.6 ps, 0.31 of the 100 ps bit period, which costs about 1 dB. The extinction-ratio penalty is $\delta_{ER} = 10 \log_{10}[(r+1)/(r-1)]$ with $r = 10^{ER/10}$:

Channel	ER (dB)	$r = P_1/P_0$	δ_{ER} (dB)
Channel 3	9.0	7.94	1.10
Channel 5	4.5	2.82	3.22

2.3 Received power, effective power and system margin

Received power per channel = launch - loss = +1 - 22.52 = **-21.52 dBm** (the same for both channels). Effective received power = $P_{rx} - \delta_D - \delta_{ER}$, and the margin = effective power - (-24 dBm).

	Channel 3 (ER 9.0 dB)	Channel 5 (ER 4.5 dB)
Received power	-21.52 dBm	-21.52 dBm
- dispersion penalty	-1.00 dB	-1.00 dB
- ER penalty	-1.10 dB	-3.22 dB
Effective received power	**-23.62 dBm**	**-25.75 dBm**
Sensitivity (BER 10^{-12})	-24.00 dBm	-24.00 dBm
System margin	**0.38 dB**	**-1.75 dB**
Requirement 3 dB	**FAILS** by 2.62 dB	**FAILS** by 4.75 dB (negative margin)

Even Channel 3 does not meet the 3 dB requirement, as the problem statement suggests. Channel 3 (ER 9.0 dB ≥ 8.2 dB) is within the validity range of the -24 dBm sensitivity, so counting its ER penalty is conservative; without it the margin would be 1.48 dB, which is still below 3 dB. For Channel 5, ER 4.5 dB is below 8.2 dB, so the -24 dBm figure is not valid and the true margin is worse than the table shows.

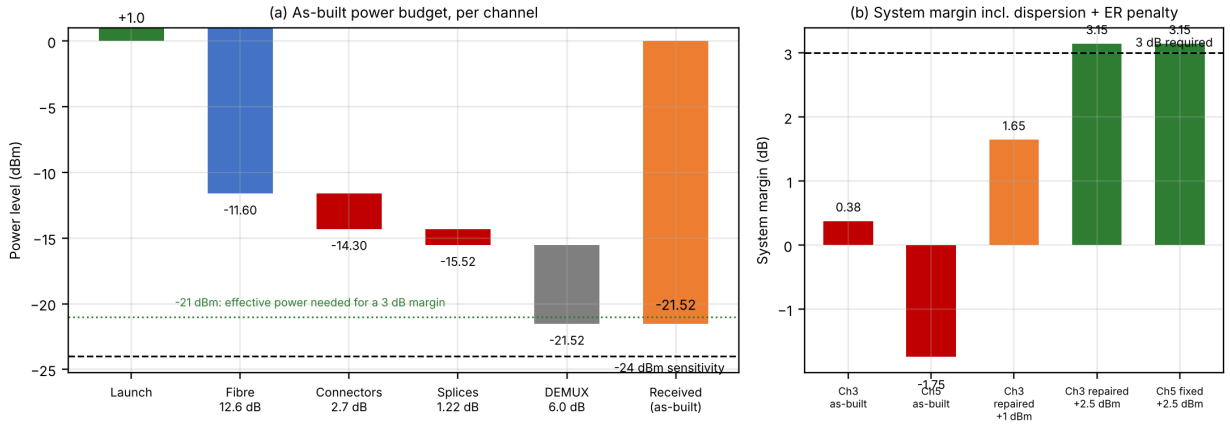


Fig. 2.1: (a) As-built power budget for one channel; (b) system margin of both channels before and after repair and launch-power change

2.4 OSNR and PMD checks

OSNR = 58 + P_{in} - NF = 58 + (-6) - 5 = **47 dB** (0.1 nm), which is 27 dB above the required 20 dB. The noise is therefore not the limit: the failure is caused by **power and penalties**. PMD: mean DGD = 0.1 × √60 = **0.77 ps**; the tolerance is 10 % of the bit period = 10 ps, so the link uses only 7.7 % of the tolerance (13× margin). PMD is negligible at 10 Gb/s (it will matter only at higher rates, where coherent DSP removes it).

2.5 Dominant excess loss and corrective action

Action	Effect on loss	Ch3 margin	Ch5 margin (ER 4.5)
As-built	22.52 dB	0.38 dB	-1.75 dB
1. Re-terminate event 3 / 4 connector (0.5 dB) and replace event 2 with a fusion splice (0.05 dB)	21.25 dB (-1.27 dB)	1.65 dB	-0.48 dB
2. Raise the booster output from +1 to +2.5 dBm per channel (+11.5 dBm total)	21.25 dB	**3.15 dB**	1.02 dB until Ch5 transmitter is replaced
3. Replace Ch5 transmitter (ER ≥ 9 dB)	-	3.15 dB	**3.15 dB**

Repairs alone reach 1.65 dB (Channel 3), still 1.35 dB short, so repair alone is not sufficient; the booster has 1.5 dB of extra capacity (input -6 dBm, +1 dBm output means 7 dB gain; +2.5 dBm needs 8.5 dB and 8 × 2.5 dBm = +11.5 dBm total, well below the EDFA saturation power of +17 dBm or more and the SBS/nonlinear thresholds of a 60 km G.652.D span at 10 Gb/s). Dispersion penalty and ER are unchanged by the power increase.

DESIGN DECISION 1 - SELECT THE PROPOSED LINK-BUDGET CONFIGURATION

I recommend that the as-built link is rejected (22.52 dB, margin 0.38 / -1.75 dB) and that the excess-loss events 3 and 2 are corrected first, followed by a launch-power increase to +2.5 dBm per channel because the budget shows excess loss (1.27 dB) and penalties, not noise, are limiting (OSNR 47 dB vs 20 dB; PMD 0.77 ps vs 10 ps): repairing the connector (-0.70 dB) and splice (-0.57 dB) recovers 1.27 dB and brings the loss back to the 21.25 dB design, and the extra +1.5 dB of launch power gives a Channel 3 margin of 3.15 dB, above the 3 dB requirement. Raising the power alone would hide the faults and leave a -22 dB reflector in the link; repairing alone is 1.35 dB short.

STAGE 3 - OTDR, SPLICE AND CONNECTOR CONFIGURATION

20 Marks | K4/K5 - Analyze and Evaluate

3.1 Pulse width, dynamic range and test arrangement

The one-way trace loss to the end of the fibre plant = 12.6 + 2.70 + 1.22 = **16.52 dB**, so the required dynamic range is 16.52 + 6 = **22.52 dB**. The pulse width must satisfy DR ≥ 22.52 dB, and the dead zones must separate the events at 59.900 and 59.909 km (9 m apart).

Pulse	DR (dB)	DR ≥ 22.52 dB?	Spatial pulse length vτ	EDZ	ADZ	Events 3/4 separate (9 m > EDZ)?	Loss of event 4 measurable (9 m > ADZ)?
10 ns	20	No (2.5 dB short)	2.04 m	1 m	4 m	Yes	**Yes**
100 ns	27	**Yes (4.5 dB spare)**	20.4 m	5 m	15 m	**Yes**	No (15 m > 9 m)
1 µs	33	Yes	204 m	40 m	100 m	No (40 m > 9 m)	No

The spatial pulse length is $v\tau = (c/n)\tau = (2.998\times10^8/1.4682)\ \tau = 2.042\times10^8$ m/s × τ. The event resolution is about half of this (1.0, 10.2 and 102 m). Events 3 and 4 are 9 m apart: **100 ns** has an EDZ of 5 m (< 9 m), so the two reflections appear as separate events, but the ADZ is 15 m (> 9 m), so the loss of the second connector cannot be measured; the 1 µs pulse merges them into one event. Only the 10 ns pulse (ADZ 4 m) can measure the loss of event 4, but its 20 dB dynamic range is too small for the full 60 km trace (22.52 dB needed).

Selected arrangement (two-step test): (1) a 100 ns, 1550 nm, long-averaging trace from the near end with a 500 m launch cable and a 500 m receive (tail) cable gives the full-link map (valid DR) and the position and loss of events 1 and 2; (2) a **10 ns trace from the far end** (the patch panel at 59.9 km becomes the start of the trace, where the full 20 dB DR is available) with a launch cable resolves events 3 and 4 and measures the loss of the second connector; (3) each splice and connector is measured **bidirectionally** (average of A-B and B-A) to remove the mode-field-diameter mismatch error, and end-faces are inspected and cleaned (IEC 61300-3-35) before every connection. The 10 ns and 100 ns results are cross-checked with the power-meter loss.

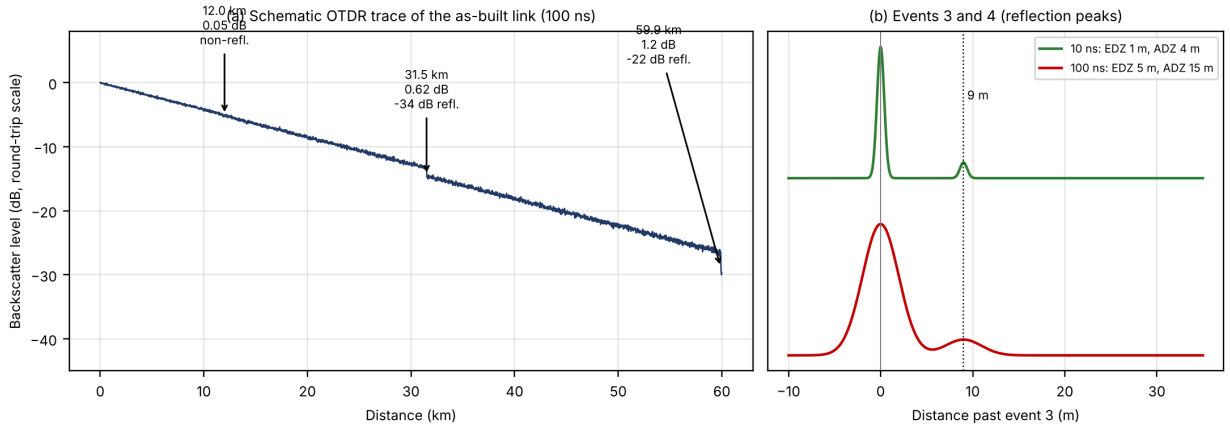


Fig. 3.1: (a) schematic OTDR trace reconstructed from the stated events (not a field capture); (b) reflection peaks of events 3 and 4 for 10 ns and 100 ns pulses

3.2 Event interpretation

Event	Position	Loss / reflectance	Interpretation	Action
1	12.00 km	0.05 dB, non-reflective	Good fusion splice (equal to the allowance)	None
2	31.50 km	0.62 dB, -34 dB	**Mechanical splice**: high loss (core misalignment, imperfect index-matching gel) and moderate reflectance; the 13th splice. Expected fusion splice: 0.02 - 0.1 dB, reflectance below -60 dB	Replace by fusion splice (0.05 dB): saves 0.57 dB
3	59.90 km	1.20 dB, -22 dB	**Defective LC/UPC connector** on the patch panel: dirty or scratched end-face, air gap or misaligned ferrule. A good UPC is 0.1 - 0.3 dB and below -50 dB; -22 dB is close to a Fresnel (air-gap) reflection and risks laser feedback	Inspect, clean, re-terminate (0.3 - 0.5 dB, below -50 dB): saves 0.70 dB or more
4	59.909 km (9 m later)	-40 dB, loss not resolved (0.5 dB by power meter)	Second connector of the patch cord (cord is 9 m long); acceptable reflectance; its loss lies inside the ADZ of the 100 ns pulse	Clean and re-measure with 10 ns pulse; keep if ≤ 0.5 dB

3.3 Fusion versus mechanical splicing

Criterion	Fusion splice	Mechanical splice
Typical loss	0.02 - 0.1 dB (0.05 dB design)	0.1 - 0.5 dB (0.62 dB measured here)
Reflectance	< -60 dB (non-reflective)	-30 to -50 dB (-34 dB measured)
Stability	Permanent, stable over temperature	Gel ageing, thermal drift
Use	Permanent plant, high-speed WDM	Emergency restoration only

3.4 Round-trip delay and location uncertainty

Round-trip delay $t = 2dn/c$, and the distance is $d = ct/(2n)$. A group-index error $\Delta n = \pm 0.002$ gives $\Delta d = d(\Delta n/n)$ with $\Delta n/n = 0.002/1.4682 =$ **0.1362 %**.

Event	Distance (km)	Round-trip delay (μs)	Location uncertainty (± m)
1	12.000	117.5	16.3
2	31.500	308.5	42.9
3	59.900	586.7	81.6
4	59.909	586.8	81.6

At event 2 the crew must search ± 43 m of cable along the route (use the cable-sheath marks, not the optical length), and at the far end ± 82 m, although the 9 m separation of events 3 and 4 is unaffected because both share the same index error (the *difference* in position is accurate to ± 0.012 m). The group index of the actual cable should be taken from the manufacturer's data and the cable-length factor (helical stranding) applied.

3.5 Connector choice for the patch-panel repair

Type	Ferrule	Insertion loss / reflectance	Suitability here
SC	2.5 mm	≤ 0.3 dB; UPC -50 dB, APC -60 dB	Larger; legacy panels
LC/UPC	1.25 mm	≤ 0.3 dB; ≤ -50 dB	**Existing panel and SFP+/CFP duplex transceivers use LC; selected**
LC/APC	1.25 mm	≤ 0.3 dB; ≤ -60 dB	Only if all connectors in the path are APC (UPC and APC must not be mated)
MPO	12-24 fibres, 2.5 mm array	≤ 0.35 dB per fibre; 0.5 dB typ.	For parallel trunks, not for this duplex pair
CS	1.25 mm, duplex, smaller	As LC	400G very-small-form-factor modules; consider for later refresh

A UPC (slightly curved, physical contact) end-face gives a low insertion loss and a reflectance of about -50 dB; an APC (8° angled) reflects the light into the cladding and gives about -60 dB, which is the better choice for analogue or reflection-sensitive systems, but it requires both sides of every mated pair to be APC. The new LC/UPC patch cords will reduce event 3 from 1.20 dB / -22 dB to about 0.3 dB / -50 dB.

3.6 Repair ranking and verification

Rank	Event	Reason
1	3 / 4 (patch panel)	Largest excess (0.70 dB), strongest reflection (-22 dB), at the transceiver end where reflections return to the laser; fastest fix (clean / re-terminate, no splice crew, same day); also protects the 100G+ upgrade
2	2 (mechanical splice at 31.5 km)	0.57 dB excess, -34 dB reflectance; needs cable access and a splice-enclosure opening at 31.5 ± 43 m, so scheduled second

Verification after repair: OTDR re-test (100 ns, bidirectional) must show no event above its allowance (splices ≤ 0.1 dB, connectors ≤ 0.5 dB, reflectance < -45 dB for the patch panel, -55 dB for splices); the power meter and stabilised source must read an end-to-end loss ≤ 21.25 dB; the end-face scope must show a PASS (IEC 61300-3-35).

DESIGN DECISION 2 - SELECT THE PROPOSED OTDR FAULT-LOCALIZATION CONFIGURATION

I recommend giving highest repair priority to the patch-panel events 3 and 4 (clean, inspect and re-terminate with LC/UPC) and then replacing the mechanical splice at 31.5 km by a fusion splice, using a 100 ns bidirectional full-link trace with 500 m launch and tail cables plus a 10 ns far-end trace for the 9 m patch cord because event 3 has the largest excess loss (0.70 dB) and the worst reflectance (-22 dB); the 100 ns pulse is the only setting with enough dynamic range (22.52 dB needed, 27 dB available) and an EDZ (5 m) shorter than the 9 m cord, while the 10 ns pulse (ADZ 4 m < 9 m) is the only one that can measure the second connector's loss; fusion splicing reduces event 2 from 0.62 dB / -34 dB to about 0.05 dB / -60 dB, and LC/UPC matches the existing equipment.

STAGE 4 - EYE-DIAGRAM, BER AND QUANTUM-LIMIT CONFIGURATION

20 Marks | K4/K5 - Analyze and Evaluate

4.1 BER from Q

BER = ½ erfc(Q/√2): Q = 7.6 → **1.48e-14**; Q = 4.7 → **1.30e-06**; Q = 7.03 → 1.03e-12

Channel	Q	BER	vs target 10^{-12}	Status
Channel 3	7.6	1.5e-14	Better by 2 decades	**Meets** (Q > 7.03)
Channel 5	4.7	1.3e-06	Worse by 6 decades	**Fails** (Q < 7.03 by 2.33)

4.2 Jitter and eye width

TJ = DJ + 14.07 RJ: Ch3 = 0.18 + 14.07 × 0.012 = **0.349 UI**; Ch5 = 0.30 + 14.07 × 0.022 = **0.610 UI**

The eye width at BER 10^{-12} = 1 - TJ = **0.651 UI** for Channel 3 and **0.390 UI** for Channel 5. **Horizontal (timing) margin:** Channel 5 still has 0.39 UI (a sampling window of 39 ps), so the eye is not closed in time but is 40 % narrower than that of Channel 3 (0.65 UI); DJ (0.30 UI, 1.7 times the Channel 3 value) dominates its jitter budget and points to bandwidth limitation / pattern-dependent distortion in the transmitter. **Vertical (noise) margin:** the ER of 4.5 dB and Q = 4.7 show that most of the failure is vertical (amplitude), since the 1/0 levels are too close and the noise cannot be tolerated.

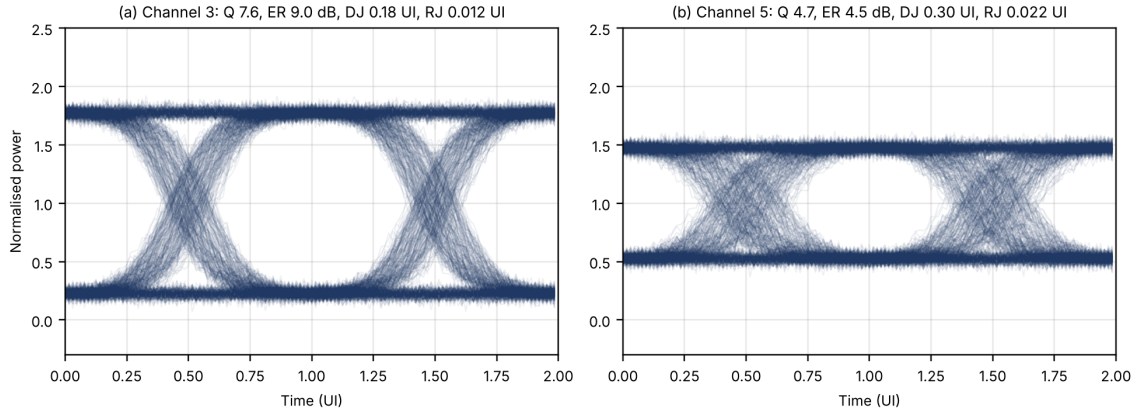


Fig. 4.1: Reconstructed eye diagrams (simulated from the stated ER, DJ, RJ and Q): Channel 3 (open) and Channel 5 (partially closed)

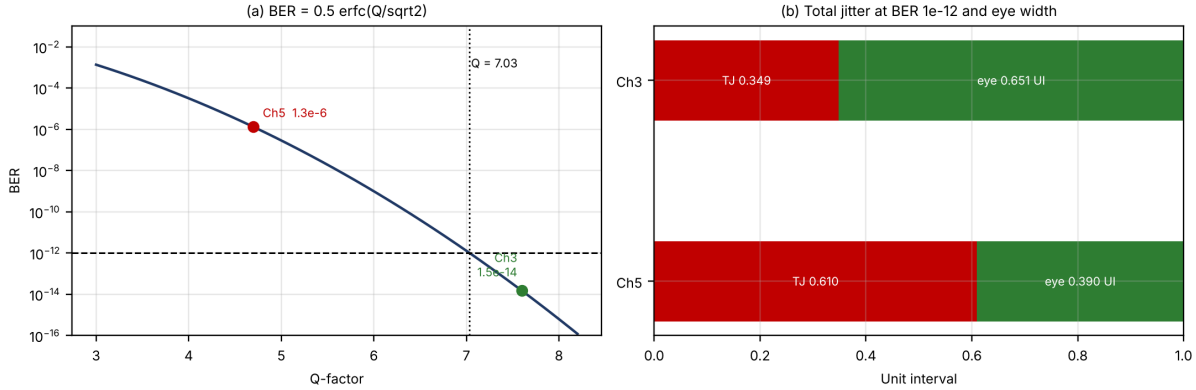


Fig. 4.2: (a) BER versus Q with both channels marked; (b) total jitter and remaining eye width for each channel

4.3 Is Channel 5's Q difference explained by its extinction ratio?

For a receiver whose noise is dominated by thermal noise ($\sigma_1 \approx \sigma_0$) at constant average power, Q is proportional to the eye amplitude ($I_1 - I_0$), i.e. to (r - 1)/(r + 1). Using r_3 = 7.94 and r_5 = 2.82:

$(r_5-1)/(r_5+1) = 0.476$; $(r_3-1)/(r_3+1) = 0.777$; ratio = 0.613; $Q_5 = 7.6 \times 0.613 =$ **4.66** ≈ measured 4.7

The ER alone predicts Q = 4.66 against the measured 4.7, an agreement of 1 %. The ER penalty of Channel 5 (3.22 dB) exceeds that of Channel 3 (1.10 dB) by 2.12 dB, which accounts for the difference in Q between the two channels. Both channels share the same fibre, connectors, splices, and dispersion penalty, so these (shared) causes affect them equally and **cannot** explain a difference between them. The root cause of the Channel 5 eye closure is therefore **channel-specific: its transmitter** (low ER from wrong bias / modulation depth or an ageing laser/modulator, plus high DJ from limited bandwidth or pattern effects). The shared fibre problem (excess loss) is a separate and additional cause of low margin for both channels.

4.4 Quantum limit for direct detection

$$BER = \tfrac{1}{2}\exp(-N_1) = 10^{-12} \rightarrow N_1 = \ln(1/(2\times10^{-12})) =$$ **26.94 photons per 1-bit**

Photons are absent in a 0-bit, so the average photon number per bit is $\bar{N} = N_1/2 =$ **13.47** photons/bit. At 1550 nm $h\nu = hc/\lambda = 1.282\text{e-}19$ J, so the sensitivity at 10 Gb/s is $P = \bar{N} h\nu B = 13.47 \times 1.282\text{e-}19 \times 10^{10} = 1.73\text{e-}08$ W = **-47.6 dBm**. The receiver sensitivity of -24 dBm is therefore **23.6 dB** (a factor of 230) from the quantum limit, because real receivers are limited by thermal noise and the amplifier (a PIN receiver needs thousands of photons per bit).

4.5 Homodyne and heterodyne PSK

Homodyne PSK: $BER = \tfrac{1}{2}\operatorname{erfc}(\sqrt{(2N)}) = 10^{-12} \rightarrow \sqrt{(2N)} = 4.974 \rightarrow N =$ **12.37 photons/bit**. Heterodyne PSK: $\sqrt{N} = 4.974 \rightarrow N =$ **24.74 photons/bit**, exactly 2 (3 dB) more than homodyne because half of the signal power is carried by the image band.

Detection	Photons per bit	Sensitivity at 10 Gb/s (dBm)	Gap from -24 dBm
Ideal direct detection (OOK)	13.47	-47.6	23.6 dB
Homodyne PSK	12.37	-48.0	24.0 dB
Heterodyne PSK	24.74	-45.0	21.0 dB

With these formulas the quantum limits of the three schemes are within 3 dB of each other (-48.0, -47.6 and -45.0 dBm), so coherent detection does not give a large quantum advantage in photons per bit at the same bit rate. Its advantage is practical: a strong local oscillator makes the receiver **shot-noise limited** instead of thermal-noise limited (the 24 dB gap largely disappears), it gives access to phase and polarisation (QPSK, 16-QAM, dual polarisation, 2×–4× the bits per symbol), and it allows digital compensation of CD and PMD.

4.6 Meaning for the coherent upgrade options

- The shot-noise limit means the required OSNR of the options (12.5, 19.5 and 26 dB for QPSK, 16-QAM and 400G) is set mainly by the constellation density, not by the receiver electronics: more bits per symbol need more photons per bit.
- Heterodyne costs 3 dB versus homodyne; modern coherent modules use intradyne detection with DSP, which obtains homodyne-like sensitivity.
- Photon-counting limits show that a 10G direct-detect receiver at -24 dBm has about 23 dB of unused sensitivity theoretically; even a perfect receiver would not recover the 1.27 dB of lost link power, so the physical repairs remain necessary.

DESIGN DECISION 3 - SELECT THE PROPOSED EYE / BER PERFORMANCE CONFIGURATION

I recommend that Channel 5 is not acceptable and its transmitter must be corrected (replace or re-bias to ER ≥ 9 dB and DJ ≤ 0.18 UI) before the channel is accepted, with Q ≥ 7.03, ER ≥ 8.2 dB, TJ ≤ 0.35 UI and BER ≤ 10^{-12} as the recovery evidence because its BER is 1.3e-06 (Q 4.7, ER 4.5 dB, which also invalidates the -24 dBm sensitivity), its eye width is only 0.39 UI, and the ER difference alone predicts its Q (4.66 vs 4.7), proving a channel-specific transmitter cause since Channel 3 shares the same fibre and meets BER 1.5e-14. The receiver is 23.6 dB from the quantum limit, so the margin must be recovered by link and transmitter repair, not by detection technology.

STAGE 5 - INTEGRATED WDM UPGRADE AND ENERGY STRATEGY

20 Marks | K5 - Evaluate

5.1 G.694.1 channel plan

$$f_n = 193.1\ \text{THz} + n \times 0.05\ \text{THz};\ \lambda = c/f$$

For **n = +3** on the 50 GHz grid: f = 193.1 + 3 × 0.05 = **193.25 THz** and λ = $2.998\times10^{8}/193.25\times10^{12}$ = **1551.36 nm**. For Option Q (32 GBd, Nyquist roll-off 0.1) the occupied bandwidth is 32 × (1 + 0.1) = 35.2 GHz; with 12.5 GHz slots this needs 35.2/12.5 = **3 slots = 37.5 GHz** per channel (a 2.816-slot width rounded up to 3). The fixed 50 GHz grid in Option P leaves 50 - 35.2 = 14.8 GHz (30 %) unused per channel.

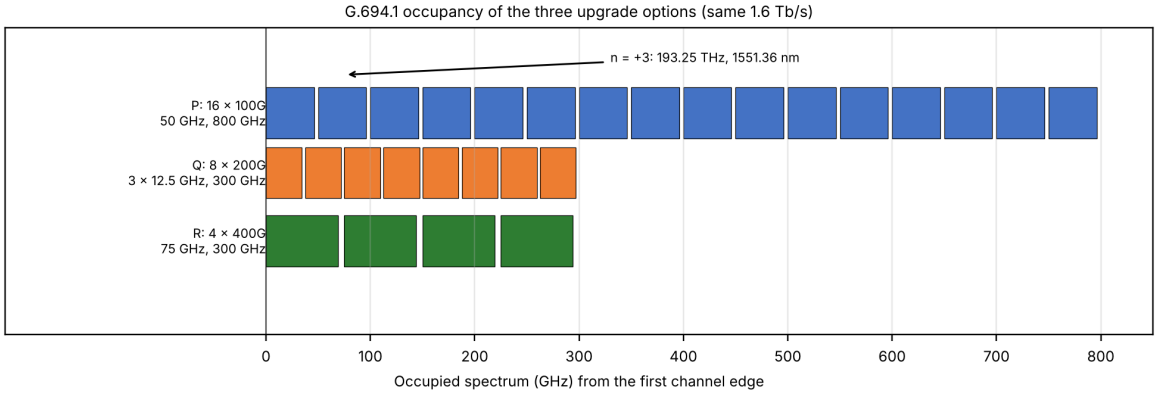


Fig. 5.1: Occupied spectrum of the three options for 1.6 Tb/s; the n = +3 channel of the 50 GHz grid is marked

5.2 Spectrum, spectral efficiency and OSNR margin

Occupied spectrum = N × slot; payload spectral efficiency = 1.6 Tb/s / occupied bandwidth. The repaired-link OSNR is 47 dB at the 8-channel booster input of -6 dBm/channel (adjusted for the channel count: 44.0 dB for 16 channels, 47.0 dB for 8 and 50.0 dB for 4 channels with the same total power).

Option	Channels	Slot	Occupied	SE (b/s/Hz)	Required OSNR	Margin (47 dB)	Margin (channel-count adjusted)
P: 16 × 100G DP-QPSK	16	50 GHz	800 GHz	2.00	12.5 dB	34.5 dB	31.5 dB
Q: 8 × 200G DP-16QAM	8	37.5 GHz (3 slots)	300 GHz	5.33	19.5 dB	27.5 dB	27.5 dB
R: 4 × 400G Si-photonic	4	75 GHz	300 GHz	5.33	26 dB	21.0 dB	24.0 dB
Legacy 160 × 10G NRZ (reference)	160	100 GHz	16 THz (does not fit)	0.10	20 dB	-	-

All three options have a large OSNR margin because the link has no in-line amplifier and so the OSNR does not degrade with distance; the margin is not the discriminating factor. The real limits are the **received power** (at +2.5 dBm the power per channel at the demultiplexer output is -18.75 dBm) and the optical front-end of the pluggables, which must be confirmed against the module data sheet. Option R has the lowest margin (21 dB) but still 21 dB above the requirement. Chromatic dispersion (1020 ps/nm) and PMD (0.77 ps) are removed by the DSP of coherent options.

5.3 Module power and energy per bit

Option	Modules	Power per module	Energy per bit	Total power for 1.6 Tb/s	vs legacy
P: 100G DP-QPSK	16	22 W	22 W / 100 Gb/s = **220 pJ/bit**	352 W	-37 %
Q: 200G DP-16QAM	8	20 W	20 W / 200 Gb/s = **100 pJ/bit**	160 W	-71 %
R: 400G Si-photonic	4	15 W	15 W / 400 Gb/s = **37.5 pJ/bit**	**60 W**	**-89 %**
Legacy 10G DWDM SFP+	160	3.5 W	3.5 W / 10 Gb/s = 350 pJ/bit	560 W	reference

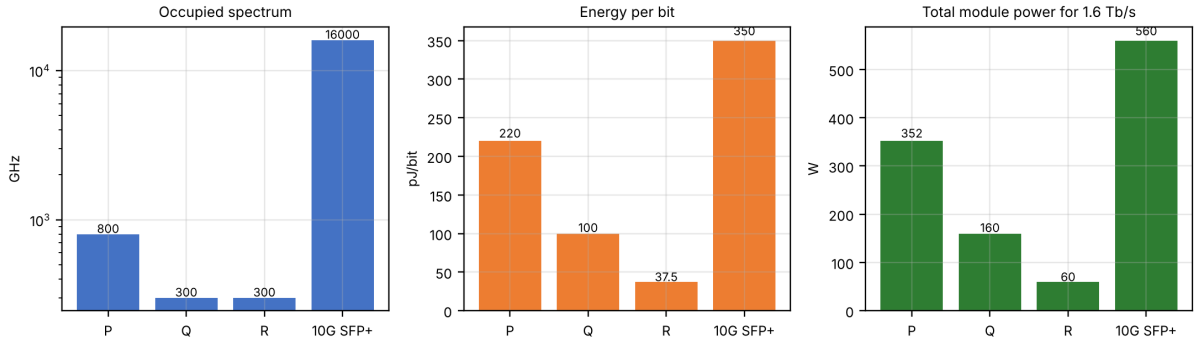


Fig. 5.2: Occupied spectrum (log scale), energy per bit and total module power of the options for 1.6 Tb/s

Role of silicon photonics. A Si-photonic engine integrates the modulators, Ge photodetectors, waveguides, couplers and thermal tuners on one CMOS-compatible chip, replacing discrete $LiNbO_3$ modulators and bulk optics; this cuts coupling and drive losses, size and cost, and lets one module carry 400 Gb/s with 15 W (37.5 pJ/bit), about 6 times better than the 100G module and 9 times better than the 10G SFP+ solution. The drawbacks are the high modulator insertion loss (compensated by amplification and DSP), temperature sensitivity (needs heaters) and a slightly lower modulator linearity.

5.4 Sequence of repair, verification and upgrade

Step	Action	Reason for this order
1	OTDR (100 ns) + power-meter baseline, end-face inspection; document the as-built loss	Defines the starting condition and gives the evidence for the repair
2	Clean / re-terminate LC/UPC connector (events 3/4); re-test	Cheapest, fastest and largest single gain (0.7 dB); removes the -22 dB reflector
3	Replace mechanical splice (event 2) by fusion splice; bidirectional OTDR re-test	Needs a field crew; second largest gain (0.57 dB)
4	Replace Channel 5 transmitter; re-measure eye, ER, Q, BER	Channel-specific fault; do not mask it with power
5	Set launch +2.5 dBm per channel; verify margin ≥ 3 dB, OSA OSNR, flatness	Last budget correction; confirms the final 3.15 dB margin
6	Install Option R (4 × 400G) on 4 grid slots; remove legacy 10G channels; re-run OTDR, OSA, BER	The upgrade can be commissioned only on a verified, low-loss, low-reflection link

5.5 Ranking and selection

Criterion (weight)	P: 16 × 100G	Q: 8 × 200G	R: 4 × 400G
Spectrum (occupied, SE)	Worst: 800 GHz, 2.0 b/s/Hz	300 GHz, 5.33 b/s/Hz (flex-grid)	300 GHz, 5.33 b/s/Hz (fixed 75 GHz)
OSNR margin (47 dB)	34.5 dB	27.5 dB	21.0 dB (still ample)
Energy per bit	220 pJ/bit	100 pJ/bit	**37.5 pJ/bit**
Modules / complexity	16 modules	8 modules	**4 modules**
Flexibility	Fixed 50 GHz	**Flex-grid 12.5 GHz**	Fixed 75 GHz (standard pluggable)
Rank	3	2	**1**

Selected: Option R. It ties with Q for the smallest spectrum (300 GHz, 5.33 b/s/Hz), has the lowest energy per bit (37.5 pJ/bit, 2.7 times less than Q), needs only 4 modules, 60 W instead of 160 W, and its 21 dB OSNR margin is more than adequate. Option Q wins only in flexibility (flex-grid), which is not a requirement here. Option P is rejected for using 800 GHz and 352 W. The 75 GHz channels leave 4.4 THz - 0.3 THz of the C-band for growth. **Measurements to repeat after the upgrade:** OTDR (loss, events, reflectance), power-meter loss and received power per channel, OSA (wavelengths on the G.694.1 grid, per-channel power and OSNR), pre-FEC BER and Q for all four channels, DSP-reported CD and PMD, coherent receiver constellation / EVM, and module power / temperature.

DESIGN DECISION 4 - CREATE THE INTEGRATED OPTIMIZATION STRATEGY

I recommend the integrated repair-and-upgrade workflow - OTDR-guided repair of events 3/4 and 2, replacement of the Channel 5 transmitter, launch power at +2.5 dBm, then installation of Option R (4 × 400G Si-photonic pluggables on 75 GHz slots) - because it delivers 1.6 Tb/s on the same fibre pair without re-laying fibre because the repaired link has a loss of 21.25 dB and a margin of 3.15 dB (≥ 3 dB), its OSNR of 47 dB exceeds the 26 dB needed by 21 dB, the upgrade occupies 300 GHz (5.33 b/s/Hz) and costs 37.5 pJ/bit (60 W against 560 W for 160 legacy 10G modules), and the diagnostic evidence ties each fault (connector, splice, transmitter) to a specific, verifiable corrective action.

STAGE 6 - VERIFY AND CRITICALLY EVALUATE THE COMPLETE LINK

10 Marks | K5/K6 - Evaluate and Create/Design

6.1 Verification table

Requirement	Specified / target	Achieved / evidence	Status
Total optical link loss	≤ 21.25 dB (design)	As-built 22.52 dB; after repair **21.25 dB** (events 2, 3 corrected; -1.27 dB)	Met after repair
Received optical power	≥ -24 dBm plus 3 dB margin (effective power ≥ -21 dBm)	As-built -21.52 dBm (effective -23.62 dBm); after repair at +2.5 dBm: -18.75 dBm (effective -20.85 dBm)	Met after repair
System margin	≥ 3 dB	Ch3 0.38 dB → 1.65 dB (repair) → **3.15 dB** (+2.5 dBm); Ch5 -1.75 dB → 3.15 dB after transmitter replacement	Met after repair
OTDR event condition	No event above allowance	Events 2 (0.62 dB, -34 dB) and 3 (1.20 dB, -22 dB) exceed; after repair: splice ≈ 0.05 dB, connector ≈ 0.3 - 0.5 dB, -50 dB; event 4 ≤ 0.5 dB (-40 dB)	Met after repair; re-test required
Eye / Q / jitter / ER	Q ≥ 7.03; ER ≥ 8.2 dB	Ch3: Q 7.6, ER 9.0 dB, TJ 0.349 UI (eye 0.65 UI) - met. Ch5: Q 4.7, ER 4.5 dB, TJ 0.610 UI (eye 0.39 UI) - fails; ER-equivalent prediction Q 4.66; transmitter replacement required	Ch3 met; Ch5 not met until Tx replaced
BER and quantum-limit gap	BER ≤ 10^{-12}	Ch3 1.5e-14 (met); Ch5 1.3e-06 (fails). Quantum limit -47.6 dBm (13.5 photons/bit); gap 23.6 dB	Ch3 met; Ch5 after Tx
Upgrade spectrum and energy per bit	1.6 Tb/s, minimum pJ/bit	Option R: 4 × 400G, 300 GHz, 5.33 b/s/Hz, 37.5 pJ/bit, 60 W; OSNR margin 21 dB	Met (selected)

Consistency with the requirement. The calculated values and measurement interpretations agree: the OTDR excess (0.70 + 0.57 = 1.27 dB) equals the difference between the as-built and design loss (22.52 - 21.25); the ER penalty explains the measured Q of Channel 5 (4.66 predicted, 4.7 measured); the BER values follow from the stated Q; the OSNR (47 dB) shows the link is power-limited, which is why repair and power, not an amplifier change, restore the margin. **Limitations:** the Q-from-ER agreement assumes thermal-noise-limited reception; the 1550 nm OTDR cannot show bend loss at other wavelengths; module power numbers are the supplied values (not measured); the margin calculation counts the ER penalty of Channel 3 conservatively, and the final 3.15 dB (4.25 dB without the ER penalty) leaves a small reserve of only 0.15 dB for future splices (a repair margin of about 0.1 dB per splice must be reserved by the operator or the launch power raised to +3 dBm).

6.2 Final system design

6.2.1 Diagnostic, repair and upgrade workflow

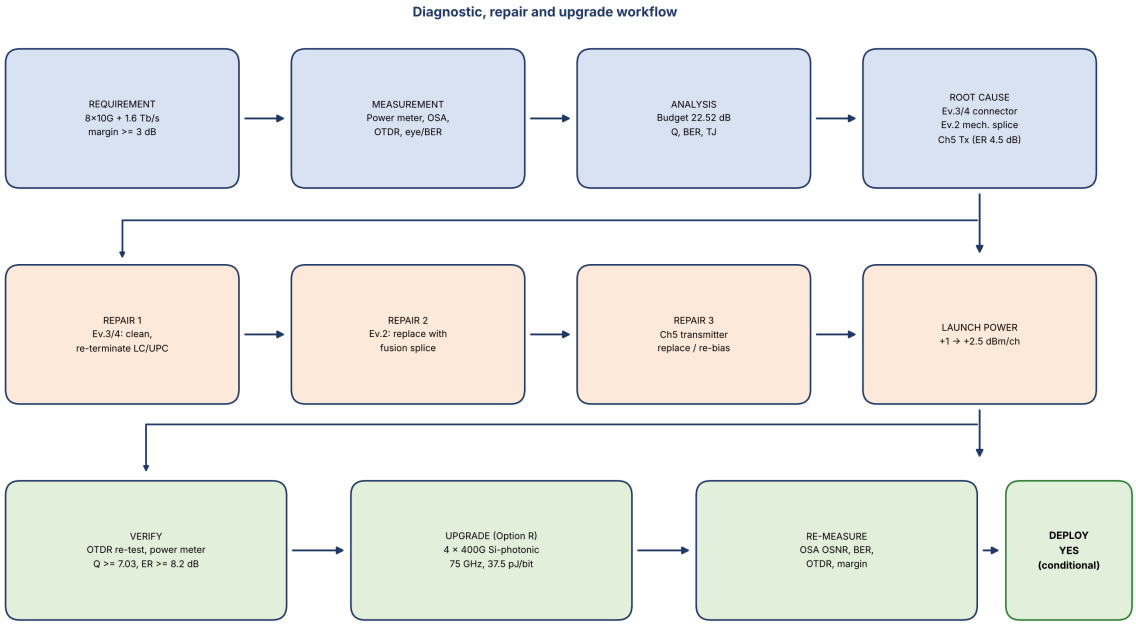


Fig. 6.1: Flowchart of the integrated diagnostic, repair and upgrade process

6.2.2 Final parameter and diagnostic table

Area	Parameter	Value adopted / decision
Link	Fibre / length / attenuation	ITU-T G.652.D, 60 km, 0.21 dB/km, n = 1.4682
Budget	Loss: design / as-built / repaired	21.25 / 22.52 / 21.25 dB
Budget	Launch per channel / received power (repaired)	+2.5 dBm (booster output +11.5 dBm total) / -18.75 dBm
Penalties	Dispersion / ER (Ch3) / ER (Ch5)	1.00 dB / 1.10 dB / 3.22 dB (Ch5 target ≤ 1.1 dB after Tx replacement)
Margin	Ch3 / Ch5 (as-built → final)	0.38 → 3.15 dB / -1.75 → 3.15 dB
Noise	OSNR / required / PMD	47 dB / 20 dB / 0.77 ps (< 10 ps)
OTDR	Pulse widths / arrangement	100 ns full link, 10 ns far-end for events 3-4; 500 m launch + tail cables, bidirectional, 1550 nm
OTDR	Events and priority	1: 12.00 km 0.05 dB OK; 3/4: 59.90/59.909 km priority 1; 2: 31.50 km (± 43 m) priority 2
Connectors / splices	Choice	LC/UPC (≤ 0.3 dB, ≤ -50 dB) at the patch panel; fusion splice at 31.5 km
Eye / BER	Ch3 / Ch5 (as measured)	Q 7.6, BER 1.5e-14 / Q 4.7, BER 1.3e-06; Ch5 target: ER ≥ 9 dB, Q ≥ 7.03
Quantum limit	Direct / homodyne / heterodyne	13.5 / 12.4 / 24.7 photons/bit; -47.6 / -48.0 / -45.0 dBm
Grid	n = +3 on 50 GHz; Option Q slots	193.25 THz, 1551.36 nm; 3 × 12.5 GHz = 37.5 GHz
Upgrade	Selected option	R: 4 × 400G Si-photonic, 75 GHz, 300 GHz, 5.33 b/s/Hz, 37.5 pJ/bit, 60 W

Final engineering decision

WOULD YOU DEPLOY THE OPTIMIZED DESIGN YOU CREATED? YES - WITH TECHNICAL JUSTIFICATION AND CONDITIONS.

- **Link budget, penalties and margin:** the as-built link (22.52 dB, margin 0.38 / -1.75 dB) is not deployable, but the repaired design reaches 21.25 dB loss and a 3.15 dB margin (4.25 dB if the ER penalty is not counted for Ch3), above the 3 dB requirement.
- **Total loss and dominant excess:** the 1.27 dB excess is the patch-panel connector (0.70 dB) and the mechanical splice (0.57 dB), both repairable without re-laying fibre.
- **OTDR diagnosis:** a 100 ns + 10 ns two-step bidirectional test, valid DR (27 dB vs 22.52 dB), EDZ 5 m and ADZ 4 m for the 9 m patch cord, locates the events within ± 43 m / ± 82 m; priority 1 is events 3/4, priority 2 is event 2.
- **Eye, jitter, ER, Q and BER:** Channel 3 meets BER 1.5e-14; Channel 5 does not (BER 1.3e-06) and its transmitter must be replaced, since the ER difference explains its Q (4.66 vs 4.7).
- **Quantum limit and coherent suitability:** the receiver is 23.6 dB from the direct-detection limit; coherent detection (shot-noise limited, DSP for CD/PMD) is therefore suitable for the upgrade.
- **Grid, spectrum and energy:** Option R uses 4 channels on 75 GHz (300 GHz, 5.33 b/s/Hz) at 37.5 pJ/bit, 60 W for 1.6 Tb/s, the best of the three options on energy and tied for the best on spectrum.
- **Reliable 1.6 Tb/s:** OSNR margin 21 dB, received power -18.75 dBm per channel, dispersion 1020 ps/nm and PMD 0.77 ps handled by the coherent DSP; the fibre is re-used.

Conditions of deployment (the YES is conditional). (1) Do not commission 1.6 Tb/s before the post-repair OTDR and power-meter tests confirm ≤ 21.25 dB, no event above its allowance and reflectance better than -45 dB at the patch panel. (2) Channel 5 must be repaired and demonstrate Q ≥ 7.03, ER ≥ 8.2 dB before it is accepted; until then it should not carry traffic that needs 10^{-12}. (3) The 3.15 dB margin is only 0.15 dB over the requirement when the ER penalty is counted; keep +2.5 dBm as the minimum launch and re-check the budget if more splices are added. (4) Check the Option R received-power and OSNR limits (26 dB) in the module data sheet, since only the figures given in the case study were used.

If the conditions are not met (redesign): if the post-repair loss exceeds 21.25 dB by more than 0.5 dB, replace the remaining connectors with low-loss LC/UPC (≤ 0.3 dB, saving up to 0.8 dB for four connectors) and raise the launch to +3.5 dBm; if Channel 5 cannot be recovered, move its traffic to a spare grid slot with a new transmitter; if any connector remains worse than -35 dB reflectance, change to LC/APC on both sides (-60 dB). Predicted effect: margin + 0.8 dB to + 1 dB, Ch5 BER to < 10^{-12}, energy per bit unchanged at 37.5 pJ/bit.

References

- ITU-T Recommendation G.652, *Characteristics of a single-mode optical fibre and cable* (category G.652.D).
- ITU-T Recommendation G.694.1, *Spectral grids for WDM applications: DWDM frequency grid*.
- IEC 61746-1, *Calibration of optical time-domain reflectometers (OTDR)*; IEC 61300-3-35, *Fibre optic interconnecting devices - Fibre optic connector endface visual and automated inspection*.
- OIF, *Implementation Agreement 400ZR* (400G DP-16QAM, 75 GHz, coherent DWDM pluggable).
- G. Keiser, *Optical Fiber Communications*, 4th ed., McGraw-Hill, 2011 (link budgets, OTDR, eye diagram, BER and Q).

- G. P. Agrawal, *Fiber-Optic Communication Systems*, 4th ed., Wiley, 2010 (dispersion penalty, extinction-ratio penalty, quantum limit and coherent detection).
- J. M. Senior, *Optical Fiber Communications: Principles and Practice*, 3rd ed., Pearson, 2009.
- K. Kikuchi, 'Fundamentals of coherent optical fiber communications,' *J. Lightwave Technol.* 34(1), 157-179, 2016.
- G. T. Reed et al., 'Silicon optical modulators,' *Nature Photonics* 4, 518-526, 2010.

CODE OF CONDUCT - I **AJAY G** (Name) / **192412161** (Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____